

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A **MOCK INTERVIEW**

<i>Criteria</i>	Technical Knowledge (2.5)	Problem Solving (2.5)	Analytical Thinking (1.5)	<i>Communication(1.5)</i>	Confidence (1)	Response to Questions(1)	<i>Total(10)</i>
<i>Weightage</i>	25%	25%	15%	15%	10%	10%	100%
<i>Q1</i>							
<i>Q2</i>							

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

Annexure A

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

TID	Items
------------	--------------

T1	Milk, Bread
----	-------------

T2	Milk, Butter
----	--------------

T3	Bread, Butter
----	---------------

T4	Milk, Bread, Butter
----	---------------------

T5	Milk, Eggs
----	------------

Question:

Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

TID	Items
------------	--------------

T1	A, B
----	------

T2	B, C
----	------

T3	A, C
----	------

T4	A, B, C
----	---------

T5	B, C
----	------

Question:

Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

TID	Items
------------	--------------

T1	A, B
----	------

T2	A
----	---

T3	B
----	---

T4	A, B
----	------

T5 A

Question:

Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4 . Dataset: Multilevel Data

TID	Items
T1	Electronics, Mobile
T2	Electronics, Laptop
T3	Electronics, Mobile
T4	Electronics, Mobile
T5	Electronics, Laptop

Question:

Explain multilevel association rules using the dataset and compare them with multidimensional

Q5. Dataset: Large Transaction Sample

TID	Items
T1	A, B, C
T2	A, B
T3	B, C
T4	A, C
T5	A, B, C

Question:

Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Q1. ANSWER:

Association rule mining finds products that are frequently purchased together. In this dataset, Milk and Bread occur together in T1 and T4, while Milk and Butter occur together in T2 and T4. Such patterns help a retailer understand customer buying behaviour and arrange products, promotions and recommendations accordingly.

Practical example: Since Milk and Bread are purchased together, the store can place them near each other or offer a combined discount. This can increase cross-selling and total sales.

Q2. ANSWER:

Apriori and FP-Growth both find frequent itemsets, but their approaches differ.

Apriori generates candidate itemsets level by level and repeatedly scans the database to calculate support. It is simple and easy to understand, but candidate generation can become expensive for large datasets.

FP-Growth avoids explicit candidate generation. It builds an FP-Tree that compresses the transaction database and mines frequent patterns from the tree.

For this small dataset, Apriori is suitable because it is simple. For large datasets with many transactions and repeated items, FP-Growth is generally preferred because it reduces candidate generation and database scans.

Q3. ANSWER:

Support measures how frequently an itemset appears in the complete transaction database:

$\text{Support}(A) = \text{transactions containing } A / \text{total transactions}$.

Here A occurs in T1, T2, T4 and T5, so $\text{support}(A) = 4/5 = 80\%$. B occurs in T1, T3 and T4, so $\text{support}(B) = 3/5 = 60\%$.

Confidence measures how often the consequent occurs when the antecedent occurs:

$\text{Confidence}(A \rightarrow B) = \text{Support}(A \text{ and } B) / \text{Support}(A)$.

A and B occur together in T1 and T4, so $\text{support}(A,B) = 2/5 = 40\%$. Therefore $\text{confidence}(A \rightarrow B) = (2/5)/(4/5) = 50\%$.

Support shows how common a pattern is, while confidence shows the reliability of an association rule.

Q4. ANSWER:

Multilevel association rules discover relationships at different levels of a concept hierarchy. In the dataset, Electronics is a higher-level category, while Mobile and Laptop are lower-level products.

For example, Electronics -> Mobile can be examined at the specific product level. Electronics occurs in all 5 transactions, while Mobile occurs in T1, T3 and T4, giving support of 60%.

Multilevel mining can reveal both general and specific patterns using the hierarchy.

Multidimensional association mining considers different attributes or dimensions such as customer age, income, product and location. Thus, multilevel focuses on abstraction levels within a hierarchy, while multidimensional combines multiple attributes.

Q5. ANSWER:

Mining large datasets can face several challenges: a very large number of candidate itemsets, repeated database scans, high memory usage, long execution time and difficulty handling rapidly growing transaction volumes.

For the sample, combinations such as {A,B}, {A,C}, {B,C} and {A,B,C} must be evaluated. In a large real dataset, the number of possible combinations can grow rapidly.

Scalable methods address these issues using techniques such as FP-Growth, which compresses transactions into an FP-Tree and avoids explicit candidate generation. Parallel or distributed mining can also divide the workload across machines. Efficient pruning, compact data structures and reduced database scans improve scalability.